Mahdi Khani

(825) 888-8888 | Edmonton, AB | mkhani@ualberta.ca | linkedin.com/mkhani97 | github.com/mkhani97

Education

University of Alberta — Edmonton, AB

Specialization BSc in Computer Science | GPA: 3.6 — *Expected April 2028*

- **Coursework:** Data Structures & Algorithms, Data Science, Statistical Analysis, Software Engineering, AI & ML, Databases, Linear Algebra, Discrete Math & Probability, Multivariate Calculus

Experience

Data Analyst Intern, IT PMO — Sep. 2025 – Dec. 2026

Bird Construction — *Edmonton, AB*

- Led migration of the PMO's entire database from SharePoint to Dataverse, designing it as a relational data model
- Manage the PMO's Dataverse environment: tables, relationships and schemas for projects, risks, issues and change requests
- Built Power BI dashboards on project health and portfolio status, presented at the executive level
- Built an end-to-end project management model-driven app in Power Apps with a four-stage lifecycle and gate approvals, extended with JavaScript, Power Fx, and Power Automate
- Automated data ingestion and approval workflows with Power Automate, including Excel-to-Dataverse imports
- Developed an MCP server connecting a Copilot Studio AI agent to Dataverse, letting PMs log status reports, risks and issues through chat

Vehicle Big Data Analyst Intern — April 2024 – April 2025

Energy Mechatronics Lab, University of Alberta — *Edmonton, AB*

- Led data processing and analysis for the Alberta Zero Emission Hydrogen Transit (AZEHT) program: 39 months of 4 Hz CAN-bus telemetry from −32 to +30°C
- Wrote a Python (pandas, NumPy) pipeline segmenting trips into urban, highway and hill routes and computing 100+ parameters per trip
- Built a Streamlit and Folium app to map routes from GPS data and generate segment-level speed, grade and energy reports
- Integrated the Open-Meteo weather API, exposing onboard temperature sensor bias of up to 18.6°C in deep cold
- Presented findings to the City of Edmonton, recommending optimal seasons, hours and locations for electric bus use

US Sports Data Analyst — Sep. 2024 – Present

Betsson Group (Sporting Solutions)

- Analyze live sports data in SQL to tune predictive betting models and deliver live odds to partner sportsbooks

Lifeguard — June 2023 – Present

City of Edmonton — *Edmonton, AB*

Leadership

Founder & President — 2026 – Present

Alberta Blockchain, University of Alberta — *Edmonton, AB*

- Founded a registered student group, authoring its constitution and leading an eight-person executive team
- Set strategy and budget, and run workshops and research projects on blockchain, DeFi and data analysis

Projects

Reinforcement Learning Air Traffic Controller | *Python, PyTorch* — Ongoing

- Training RL agents to prevent near-misses and optimize aircraft sequencing for taxiing, takeoffs and landings

Algorithmic Crypto Trading Bot | *Pine Script, Predictive Analytics* — 2025

- Built an autonomous trading system that generated $200K+ in 2025, then retired it once its market edge faded

Awards & Honours

Undergraduate Leadership Award | Jason Lang Scholarship (2x) | Dean's Honour List (2x) | Top 10 U20, Edmonton Marathon